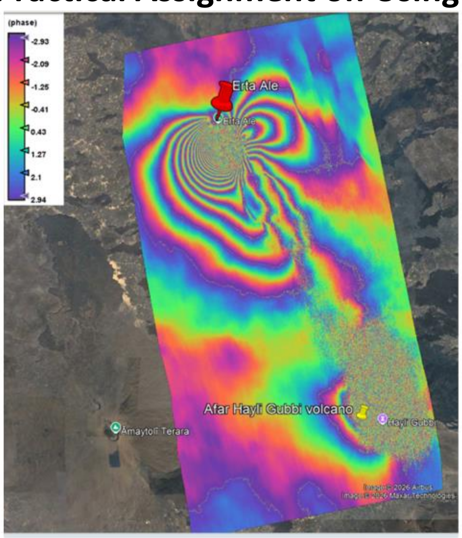
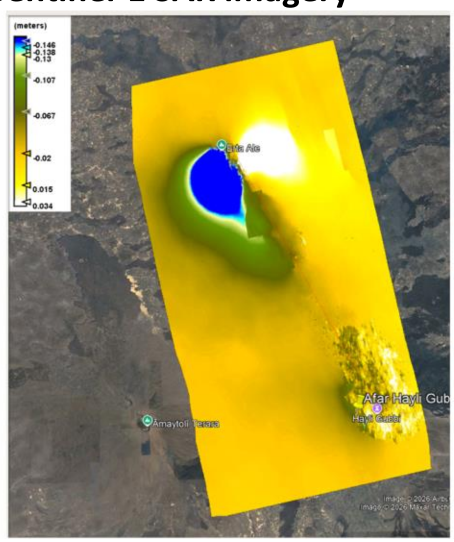
Bahir Dar University

Department of Geography and Environmental studies

Masters in Geo-Information Science

Advanced Remote Sensing Course
Practical Assignment on Using Sentinel-1 SAR imagery





Lab 2: SAR Interferometry(InSAR) for Surface Deformation analysis Using SNAP 13.

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Submitted to: Dr. Daniel Ayalew
Date: May 20, 2026

1. Interferometric SAR(InSAR) Lab Mansual

The goal of this manual is to provide students users with step-by-step instructions on interferometric processing with Sentinel-1 Interferometric Wide Swath products.

The data used in this tutorial includes the Afar area, Ethiopia, where a considerable number of volcanic activities and surface movement is reported in the region.Particularly it studies eruption activity that occurred on November 23, 2025.Using a pair of SAR SLC data and InSAR techniques this tutorial develops interferogram that identify the twin volcanic craters involved, measure the surface deformation and displacement occurred during the stated time.

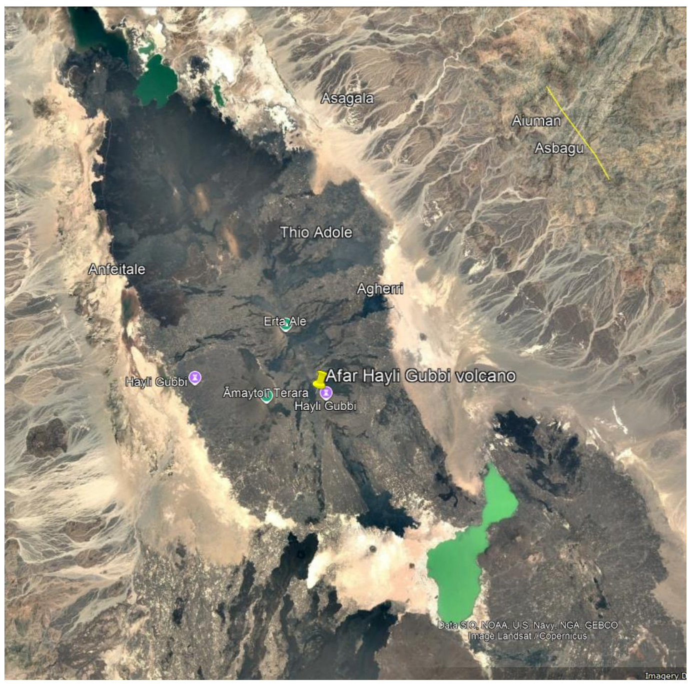


Figure 1 Study area includes the Afar,Guba Haili, Afdera Lake, Ethiopia

1.1.InSAR Background

Interferometric synthetic aperture radar (InSAR) exploits the phase difference between two complex radar SAR observations taken from slightly different sensor positions and extracts information about the earth’s surface movement.

A SAR signal contains amplitude and phase information. The amplitude is the strength of the radar response and the phase is the fraction of one complete sine wave cycle (a single SAR wavelength). The phase of the SAR image is determined primarily by the distance between the satellite antenna and the ground targets.

By combining the phase of these two images after coregistration, an interferogram can be generated whose phase is highly correlated to the terrain topography. In the case of differential interferometry (DInSAR), this topographic phase contribution is removed using a digital elevation model (DEM). The remaining variation in the interferogram can be attributed to surface changes which occurred between the two image acquisition dates, as well as unwanted atmospheric effects.

1.2.Sentinel-1 Interferometric Wide(IW) Swath Products

The Interferometric Wide (IW) swath mode is the main acquisition mode over land for Sentinel-1. It acquires data with a 250 km swath at 5 m by 20 m spatial resolution (single look). IW mode captures three sub- swaths using Terrain Observation with Progressive Scans SAR (TOPSAR). With the TOPSAR technique, in addition to steering the beam in range as in ScanSAR, the beam is also electronically steered from backward to forward in the azimuth direction for each burst, avoiding scalloping and resulting in homogeneous image quality throughout the swath.

IW SLC products consist of three sub-swaths (Figure 1, red labels) image per polarization channel, for a total of three (single polarizations) or six (dual polarization) images in an IW product. Each sub-swath image consists of a series of bursts (Figure 1, white labels), where each burst has been processed as a separate SLC image. The individually focused complex burst images are included, in azimuth-time order, into a single sub-swath image with black-fill demarcation in between, similar to ENVISAT ASAR Wide ScanSAR SLC products.

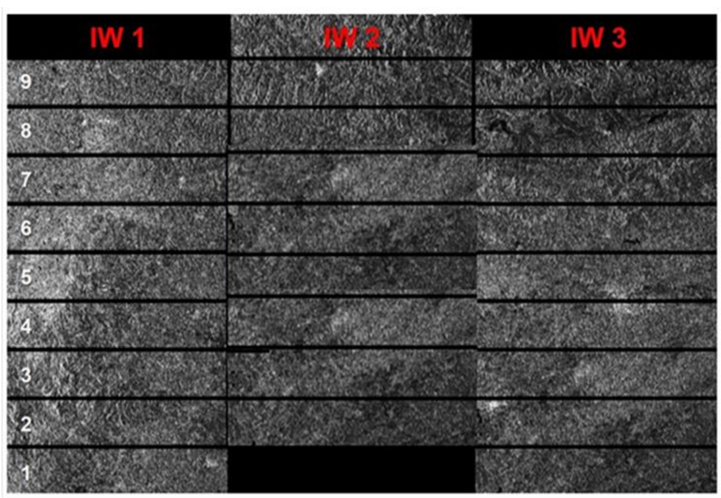


Figure 1:Sub-swaths (red) and bursts (white numbers) of S1 SLC IW products.

Source[4]: (ESA)

1.3.Data Preparation

The two datasets for this lab will be shared as **SAR_Lab_2** data in zipped folder, which contain

- S1C_IW_SLC__1SDV_**20251119**T152635_20251119T152711_005085_00A12F_E1B8
- S1A_IW_SLC__1SDV_**20251125**T152734_20251125T152800_062036_07C2C4_0277

As shown in the dataset names, the dates the two SLC datasets captured by sentinel 1 are stated as 20251119 (November 19, 2025) and 20251125(November 25,2025).

The datasets can also downloaded from the Copernicus Open Access Hub at: https://scihub.copernicus.eu/dhus (login required, registration is free) following the similar steps in lab 1 manual by adjusting the study area search to Afar Guba Hailli area, product type to SLC and filtering dates to (November 19, 2025) and (November 25, 2025). These details can be obtained from the image name itself by understanding Sentinel-1 Product Naming Convention as shown below.

Breakdown of our Specific File Name:
S1C_IW_SLC__1SDV_20251119T152635_20251119T152711_005085_00A12F_E1B8

- **S1C**: **Mission Identifier**. Refers to the Sentinel-1C satellite.
- **IW**: **Mode/Beam Identifier**. Stands for *Interferometric Wide* swath mode.
- **SLC**: **Product Type**. Stands for *Single Look Complex* data.
- **_**: **Resolution Class**. Underscore is used when resolution class does not apply to the specific product type.
- **1**: **Processing Level**. This is a *Level-1* data product.
- **S**: **Product Class**. Denotes *Standard* operational data.
- **DV**: **Polarisation**. Dual-polarisation, specifically *Dual VV + VH*.
- **20251119T152635**: **Start Date and Time**. Acquired on **November 19, 2025**, at 15:26:35 UTC.
- **20251119T152711**: **Stop Date and Time**. Scene collection ended on **November 19, 2025, at 15:27:11 UTC.**
- **005085**: **Absolute Orbit Number**. The specific orbital loop count since launch.
- **00A12F**: **Mission Data Take ID**. Unique ID for the specific continuous data stream.
- **E1B8**: **Product Unique Identifier**. A 4-character hex code used to ensure every ZIP/SAFE file has a completely unique folder name.

1.4.Software

For this Lab, we use the Open source ESA SNAP 13 (Sentinel Application Platform). It can be downloaded from https://step.esa.int/main/download/snap-download/.

1.1.1. Install SNAPHU

In addition to SNAP an external free software Plug in, SNAPHU (statistical- cost network-flow algorithm for phase unwrapping) has to be installed separately.
To install the plugin in SNAP, go to Plugins (under the Tools menu, Figure 1) and search for "SNAPHU Unwrapping" in the Available Plugins tab. Select it and click Install.

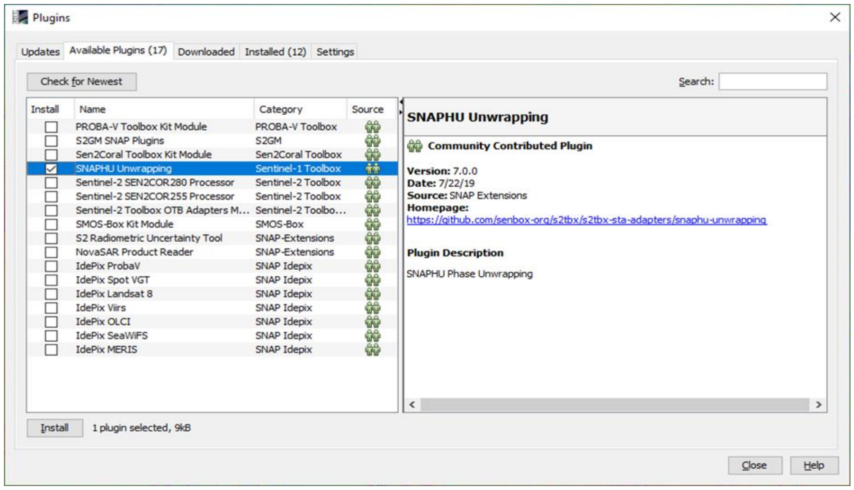


Figure 2 The Plugins interface

You will be asked to proceed with “Next” and restart SNAP. This creates a menu entry for snaphu unwrapping in the Interferometry menu.

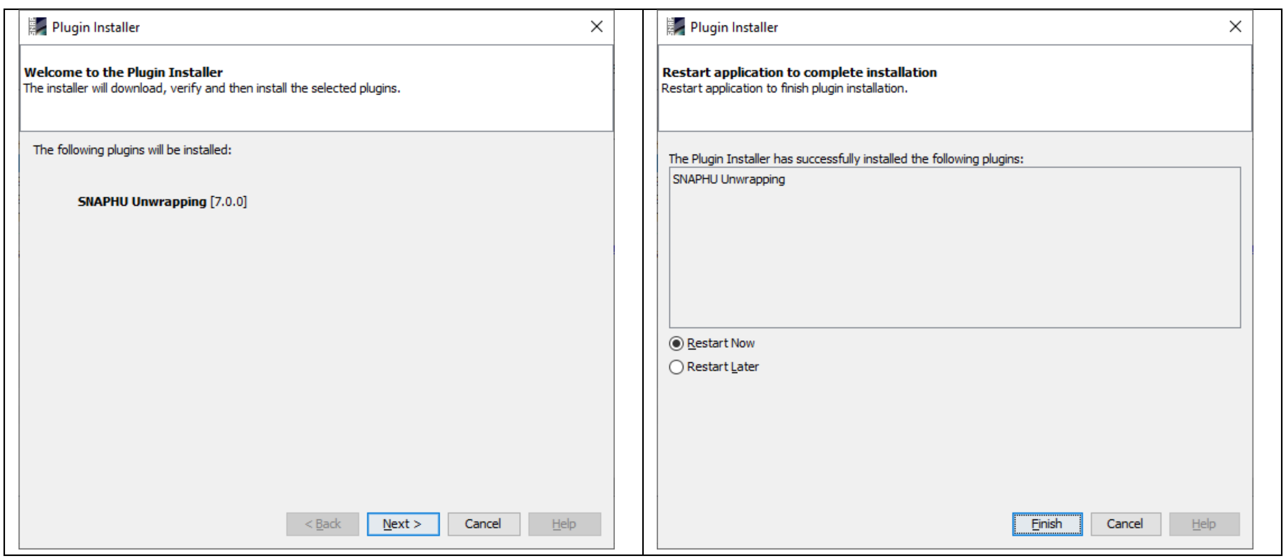


Figure 3: Installation of the snaphu plugin

After restarting SNAP, you need to install the snaphu bundles as well. Go to Manage External Tools in the Tools menu. Select “Snaphu-unwrapping” and click Edit the selected operator to open the configuration (Figure 5). Select a suitable Target Folder (here C:\GIS) where snaphu will be installed and proceed with Download and Install Now. You will then get a confirmation “Bundle was installed in location C:\GIS\snaphu-v8.0.0_win64” (this may differ according to version, path and operating system).

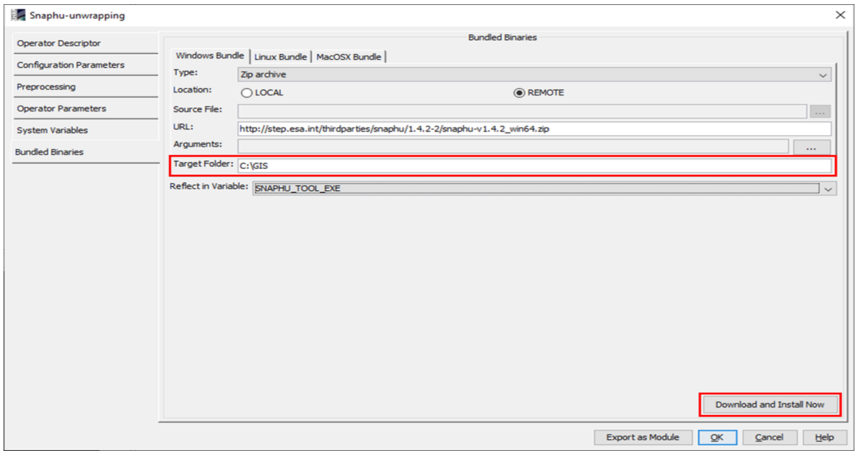


Figure 4: Installation of the snaphu bundles

Open the SAR SLC Products in SNAP

Use the Open Product button in the top toolbar and browse for the location of the downloaded data. Select the two zip files and press Open Product.

In the Products View you will see the opened products. Each Sentinel-1 product consists of Metadata, Vector Data, Tie-Point Grids, Quicklooks and Bands (which contains the actual raster data, organized by polarization and sub- swath as shown in Figure 5).

As the data is still compressed in the zip file and each sub-swath covers a large area, it is not advised to open all the bands in their current state, because it takes a considerable time until SNAP loads the raster. If you still want to check the data before proceeding double-click on the Intensity_IW1_VV band to view the raster data.

To identify the location of your study area within the product (required for the next step), you can use the World View or World Map (to see its full extent on a base map) or open the Quicklook for a preview of the dataset in an RGB color representation.

The World View or World Map menus are also a good way to check if the two images belong to the same relative orbit (required for interferometry) and if they are shifted along-track.

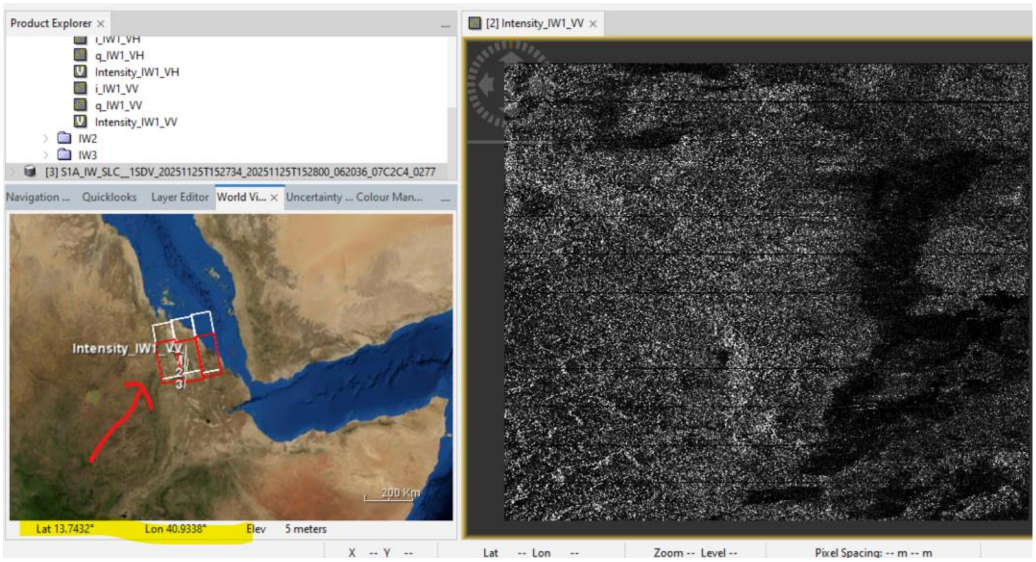


Figure 5: Product Explorer and World Map. By zomming in the world view ,you can see our study area(Afar Guba Haili)is contained in Sub swath 2.

2. SAR Interferometry Workflow

2.1.TOPS Split

S-1 TOPS Split is applied to the data to select only those bursts which are required for the analysis. To reduce the loaded data to our area of interest,

- Open the S-1 TOPS Split operator (under Radar > Sentinel-1 TOPS) and
- Select sub-swath IW2 and VV polarization in the Processing Parameters.
- Then reduce the bursts to 1 to 5 by dragging the grey triangles towards the middle(Figure 6).
- Use the preview to check the location of the selected area.
- Confirm with Run and
- Repeat the step for the second image.

Note: If the area of interest covers several Sub-Swaths, they have to be merged with the S-1 TOPS Merge operator after the Sub-Swaths were debursted.

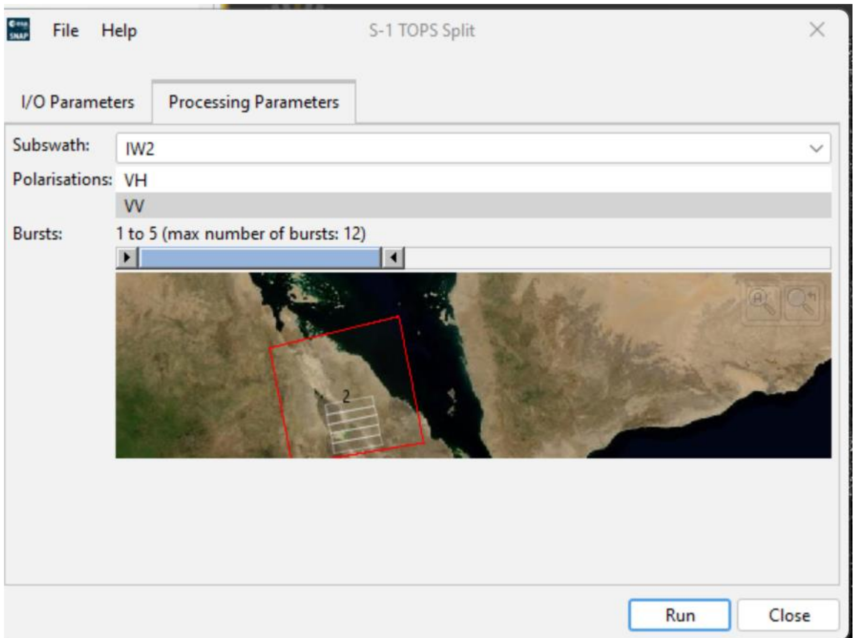


Figure 6: TOPS Split By zomming in the map ,
you can see our study area(Afar Guba Haili) is contained in bursts 1 to 5

2.2.Applying Orbit Information

Orbit auxiliary data contain information about the position of the satellite during the acquisition of SAR data. They are automatically downloaded for Sentinel-1 products by SNAP and added to its metadata with the Apply Orbit File (under the menu point Radar) operator.
The Precise Orbit Determination (POD) service for Sentinel-1 provides restituted orbit files and precise Orbit Ephemerides (POE) orbit files. POE files cover ca. 28 hours and contain orbit state vectors at intervals of 10 seconds intervals. Files are generated one file per day and are delivered within 20 days after data acquisition.

If precise orbits are not yet available for your product, restituted orbits can be selected which may not be as accurate as the Precise orbits but will be better than the predicted orbits available within the product.

- Radar > Apply Orbit File
- Execute the operator for both split products as generated in the previous step.

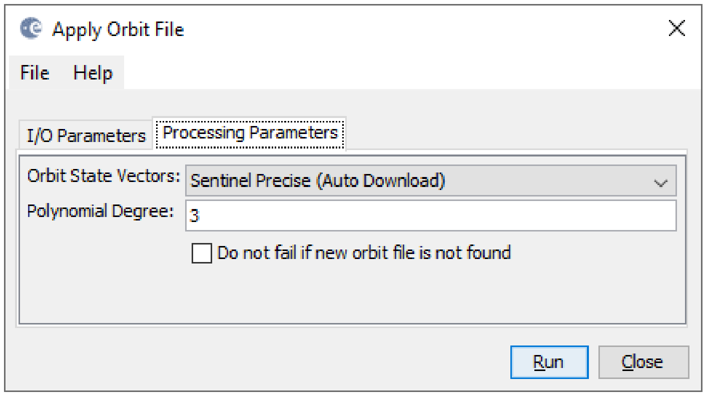


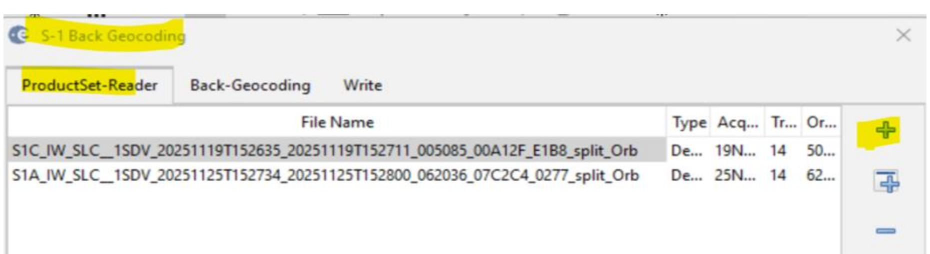
Figure 7 Apply Orbit File

2.3.Coregistration (Back Geocoding) and Enhanced Spectral Diversity

The S-1 Back Geocoding operator coregisters the two split products based on the orbit information added in the previous step and information from a digital elevation model (DEM) which is downloaded by SNAP.

- Radar > Coregistration > Sentinel-1 TOPS Coregistration
- Add the products of the two dates (ending with "_split_Orb") to the file list in the ProductSet-Reader tab and
- Select SRTM 1Sec HGT (AutoDownload) in the Back-Geocoding tab (Figure 8).

Note: Please be aware that SRTM is only available between 60° North and 54° South. Ethiopia is coered in this regeon. However, in case your area lies outside this coverage (https://www2.jpl.nasa.gov/srtm/coverage.html), you can use one of the other DEMs with AutoDownload option or use an external DEM (make sure it stored as a GeoTiff and projected in geographic coordinates / WGS84).



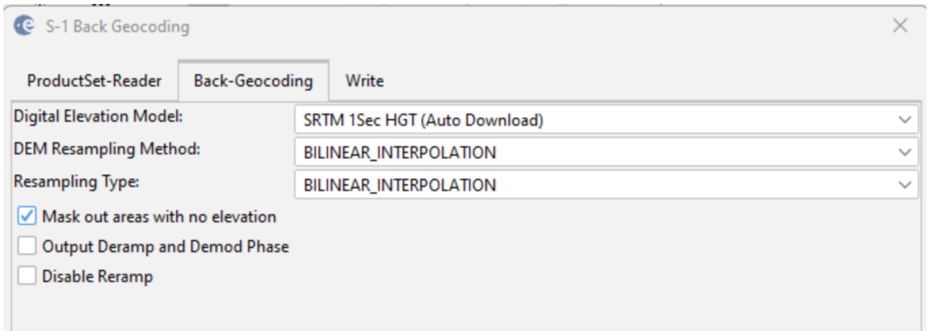


Figure 8: Back Geocoding

To increase the quality of the coregistration you apply the S-1 Enhanced Spectal Diversity (ESD) operator on the stack generated by the Back Geocoding. It applies range and azimuth shift corrections to the secondary image.

- Under Radar > Coregistration > Sentinel-1 TOPS Coregistration
- Select the stack generated by Back Geocoding
- Note: This operator is only needed when you select more than one burst. If you only selected one burst in the TOPS Split step, you do not need the ESD operator.

3. Interferogram Formation and Coherence Estimation

An interferogram is formed by cross multiplying the reference image with the complex conjugate of the secondary image. The amplitude of both images is multiplied while the phase represents the phase difference between the two images.

The interferometric phase of each SAR image pixel would depend only on the difference in the travel paths from each of the two SARs to the considered resolution cell. Accordingly, the computed interferogram contains phase variation (ϕ) from several contributing factors, most importantly the flat-earth phase ϕflat (earth curvature), the topographic phase ϕDEM (topographic surface of the earth), atmospheric conditions ϕatm (humidity, temperature and pressure change between the two acquisitions), and other noise ϕnoise (change of the scatterers, different look angle, and volume scattering), and lastly the eventual surface deformation ϕdisp which occurred between the two acquisitions (Equation 1)[2]. Contributors to SAR interferometric phase[2].

$$\phi = \phi DEM + \phi flat + \phi disp + \phi atm + \phi noise \qquad (1)$$

Differential SAR interferometry tries to estimate the contribution from the earth's surface (ϕflat and ϕDEM) which is considered equal for both image acquisitions and remove them from the interferogram so that the remaining phase variation can be attributed to surface elevation changes between both image acquisitions (Equation 2)[2].This works best if atmospheric contributions and other noise is kept as small as possible because they are hard to model. It is therefore advisable to use images from the dry season and with small perpendicular baseline.

$$\phi disp + \phi atm + \phi noise = \phi - \phi DEM - \phi flat \qquad (2)$$

Besides the interferometric phase, the coherence between the reference and the secondary image is estimated as an indicator for the quality of the phase information. Basically, it shows if the images have strong similarities and are therefore usable for interferometric processing. Loss of coherence can produce poor interferometric results and is caused by temporal (over vegetation and water bodies), geometric (errors or inaccuracies in the orbit metadata) and volumetric decorrelation (potential scattering mechanisms of voluminous structures, such as complex vegetation or dry surfaces).

NOTE: accordingly, repeat-pass interferometry (phase difference of images of two different acquisition dates) will rarely lead to usable results over dense vegetation, especially with C-band SAR data of Sentinel-1 which has a wavelength of around 5.5 cm).

Coherence is calculated as a separate raster band and shows how similar each pixel is between the secondary and reference images in a scale from 0 to 1. Areas of high coherence will appear bright. Areas with poor coherence will be dark. In the image, vegetation is shown as having poor coherence and buildings have very high coherence.

To compute the interferogram and the coherence bands,

- Radar > Interferometric > Products>Apply the Interferogram Formation operator
- Select the stack you generated after Back Geocoding and ESD as an input and
- Check the boxes: Subtract flat-earth phase:
    - The flat-earth phase is the phase present in the interferometric signal due to the curvature of the reference surface. The flat-earth phase is estimated using the orbital metadata information and subtracted from the complex interferogram.
- Check Subtract topographic phase:
    - Flat terrain should produce a series of regularly spaced, parallel fringes. Any deviation from a parallel fringe pattern can be interpreted as topographic variation. If you want this topographic variation (e.g. for the derivation of a digital elevation model), you do not subtract the topographic phase. As this workflow is interested in the surface displacement (Equation 2), topographic phase removal is applied by selecting the SRTM 1Sec HGT (Auto Download).

- Check the Include coherence estimation box: This produces a coherence band in the output calculated based on a window of 10x3 pixels in range/azimuth direction.

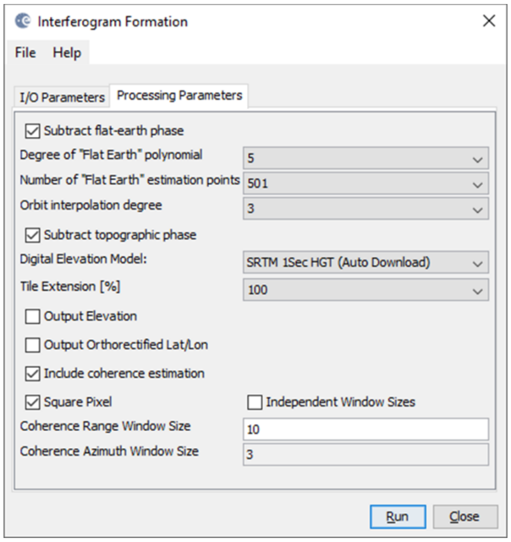


Figure 13: Interferogram Formation

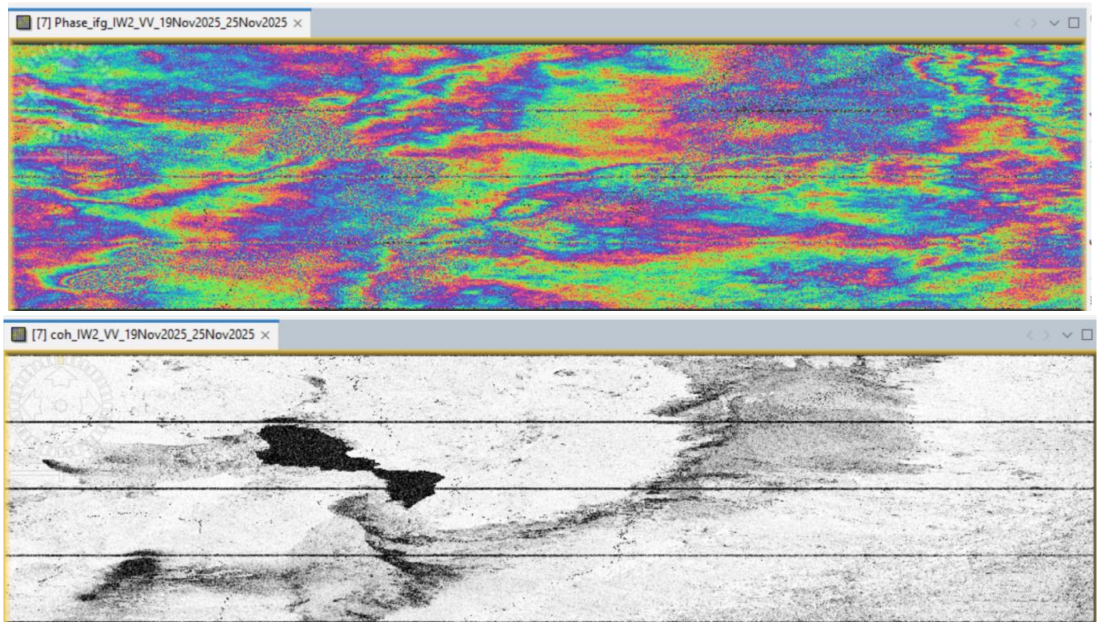


Figure 14: Interferogram (top) and coherence (bottom)

The resulting product will contain the interferogram (Phase_ifg_IW…) and the coherence (coh_IW_...) as separate bands. Double click them to check for their quality. Both products still contain the black seamlines separating the single bursts. They will be removed in the following step.

As topographic phase removal was applied, the interferogram should now only contain variations from displacement, atmosphere and noise.

- The interferogram is displayed in a rainbow color scale ranging from -π to +π.
- The patterns, also called "fringes" represent a full 2π cycle and appear in an interferogram as cycles of arbitrary colors, with each cycle representing half the sensor's wavelength.
- **Relative ground movement between two points is later derived by counting the fringes and multiplying by half of the wavelength. The closer the fringes are together, the greater the strain on the ground.**
- The coherence image shows the areas where the phase information is coherent, which means that it can be used to measure deformation (as in this tutorial) or topography (without removing the topographic phase).

While urban areas and agricultural land are displayed in white and show high coherence values (above 0.6), forest areas are dark and have low coherence (below 0.3). Accordingly, phase information over forest areas is not usable here. If low coherence areas are too dominant in the image, the later unwrapping will fail and produce faulty or random results.

3.1.TOPS Deburst

To remove the seamlines between the single bursts,

- The S-1 TOPS Deburst operator (under Radar > Sentinel-1 TOPS) is applied to the interferogram product.
- It does not require any user input (Figure 16).
- The output contains the same bands as the input, but with merged bursts according to their zero Doppler time.

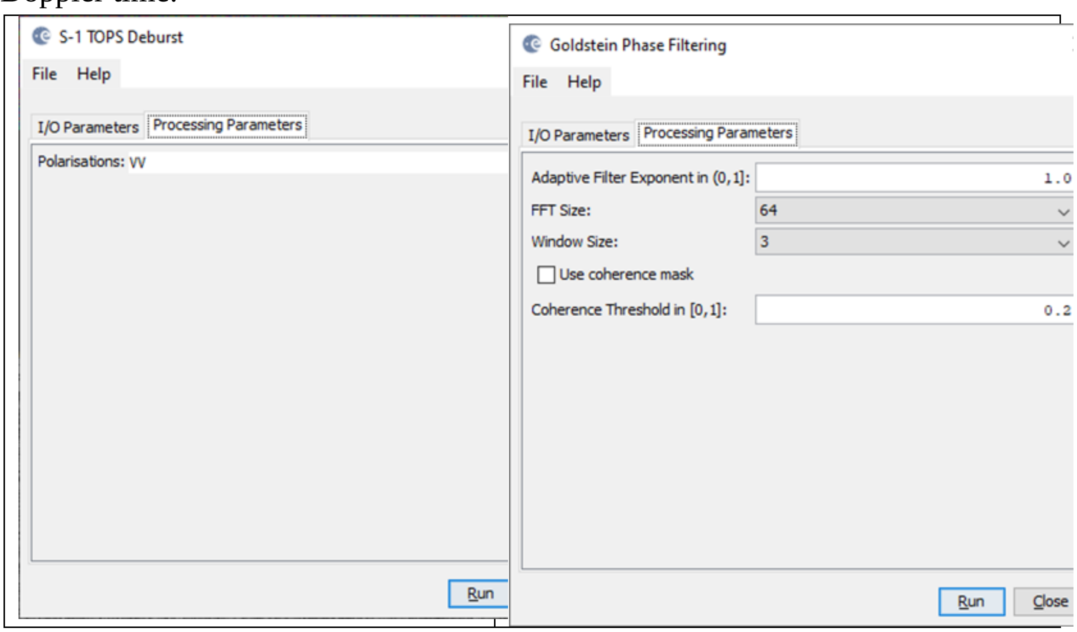


Figure 16: TOPS Deburst (left) and Goldstein phase filtering (right)

3.2. Goldstein Phase Filtering

Interferometric phase can be corrupted by noise from temporal and geometric decorrelation, volume scattering, and other processing errors. Phase information in decorrelated areas cannot be restored, but the quality of the fringes existing in the interferogram can be increased by applying specialized phase filters, such as the Goldstein filter which uses a Fast Fourier Transformation (FFT) to enhance the signal-to-noise ratio of the image. This is required for a proper unwrapping in the subsequent step. A detailed description of this filter and its parameters is given in the publication of Goldstein & Werner (1998).

- Apply the Goldstein Phase Filtering (under Radar > Interferometric > Filtering) to the debursted image. The output is a filtered phase image, the coherence is not affected.

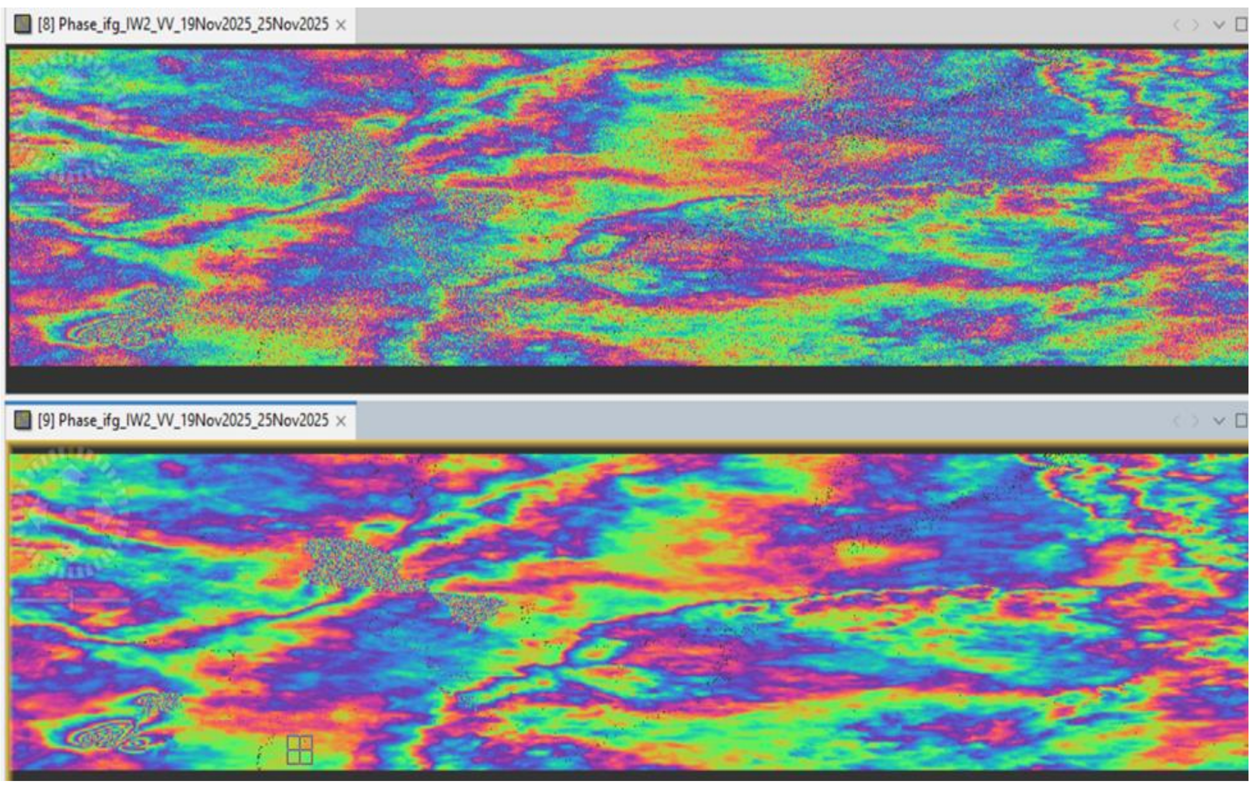


Figure 17: Interferogram before (left) and after (right) Goldstein phase filtering

3.3. Subset

As the selected bursts cover a larger area than required for this analysis, a subset can be generated. Please note that the subset should only be applied on a S1 SLC product after TOPS Deburst. Technically, the subset could have been applied before the phase filtering, but sometimes patterns become visible after applying the filter and it is good to have a look at the filtered result before selecting the area of interest.

Furthermore, the subset allows to get rid of the green areas at the edges of the image which were introduced during the processing.

To create a subset,

- Select the filtered product in the Product Explorer and
- Raster >Subset. Enter the following Pixel coordinates
- Scene start X:1064
- Scene start Y:4018
- Scene end X:3390
- Scene end Y:5263

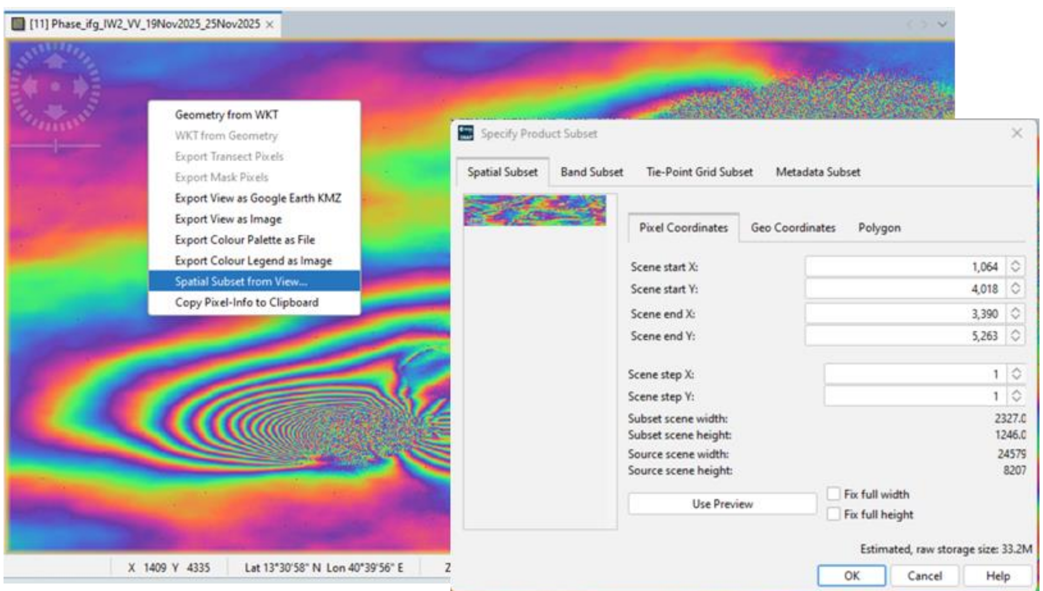


Figure 18: Subset

It is also possible to enter geographic coordinates, but as the product is still stored in slant range geometry, these are not very precise. Furthermore, it is possible to draw a rectangle in the image preview.

A new product is show in the Product Explorer, starting with ***subset_0_of_*.** This product is only of temporary nature and lost if SNAP is closed.

- To permanently store it, right-click on the product and select Save Product As… and save it in BEAM DIMAP format under the name
- 20251119_20251125_split_Orb_Stack_esd_ifg_deb_flt_subset.dim

Note: for reasons of consistency, the suffix subset is placed at the end of the filename.

3.4.Phase Unwrapping

In the interferogram, the interferometric phase is ambiguous and only known within the scale of 2π. To be able to relate the interferometric phase to the topographic height, the phase must first be unwrapped. The altitude of ambiguity is defined as the altitude difference that generates an interferometric phase change of 2π after interferogram flattening.

Phase unwrapping solves this ambiguity by integrating phase difference between neighboring pixels. After deleting any integer number of altitudes of ambiguity (equivalent to an integer number of 2π phase cycles), the phase variation between two points on the flattened interferogram provides a measurement of the actual altitude variation (Figure 19). Accordingly, unwrapped results should be interpreted as a relative height/displacement between pixels of two images.

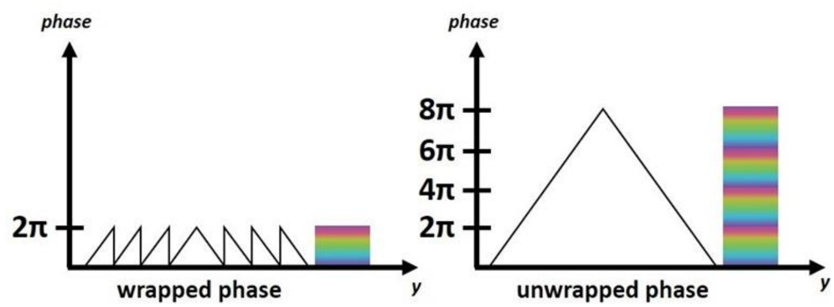


Figure 19: Principle of phase unwrapping.Source:ESA,[2]

For optimal unwrapping results it is recommended to filter the interferogram as done in the previous step and eventually apply multi-looking (under Raster > SAR Utilities). As the fringe patterns of the surface displacement are quite fine scaled (Figure 17), no multi-looking is applied in this case. For the derivation of large-scale displacement maps, multi-looking to a spatial resolution of 20 meters or lower should be considered.

The quality and reliability of unwrapped results very much depends on the input coherence. Reliable results can only be expected in areas of high coherence. Although no definite threshold exists, a minimum coherence of 0.3 is suggested.

Unwrapping in SNAP follows three distinct steps:

- Export of the wrapped phase (and definition of the parameters)
- Unwrapping of the phase (performed outside SNAP by SNAPHU)
- Import of the unwrapped phase back into SNAP

Because snaphu is an independent software (see under the section "Install snaphu" at the beginning of this document), the unwrapping is executed outside SNAP, but the plugin allows to call this execution, so no command line processing is required any longer.
Depending on the capabilities of the system, the unwrapping can take a considerable amount of time. However, snaphu supports multi-threading which means that the computation can be distributed over multiple processor cores. This has to be defined during the export.

3.4.1. Export

- The Export operator (under Radar > Interferometric > Unwrapping) converts the interferogram (as the wrapped phase) into a format which can be read by snaphu. Furthermore, a couple of parameters can be selected which affect the unwrapping process and stored in a configuration file which is accessed by snaphu.
- The source product is the subset created in the previous step. In the SnaphuExport tab, select your working directory as the target folder (e.g: D:\afar\SNAPLab\unwrap_Export). If the selection of the directory does not work, simply copy and paste the path of your working directory into the text field.
- Select DEFO as Statistical-cost mode
- Note: You can neglect the eventual error message (Error: [NodeId: SnaphuExport Please add a target folder]. It will go away once you switch tabs or hit Validate or Run, as long as you have entered a valid target folder.
- To learn about the meaning and impact of all parameters, please consult the official documentation.

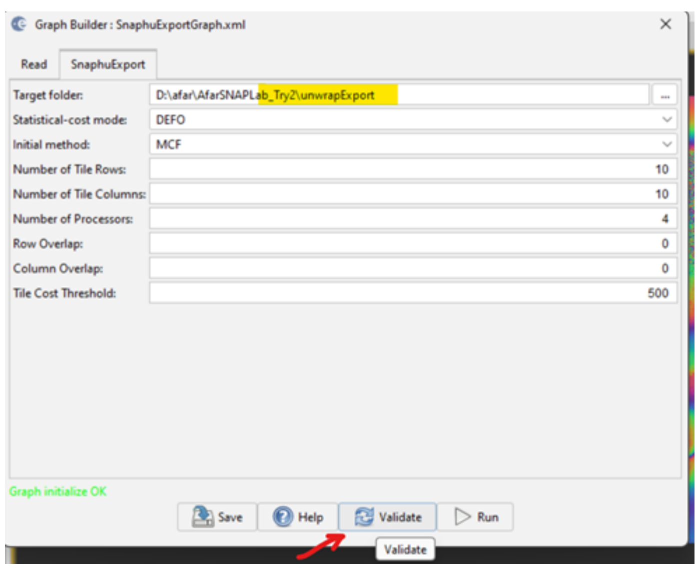


Figure 20: Snaphu export

- Hit Run to start the export.
- A directory is created in your working directory with the same name as the product you selected as an input (20251119_20251125_split_Orb_Stack_esd_ifg_deb_flt_subset).
- Inside this folder, you will find the following four files created:
  - The coherence: image (*.img) and metadata (*.hdr)
  - The wrapped phase: image (*.img) and metadata (*.hdr)
  - The unwrapped phase: only the metadata (*.hdr), because the image (*.img) is first to be created by snaphu in the next step.
  - A configuration file (snaphu.conf) containing the parameters defined in the export operator

3.4.2. Unwrapping with snaphu

- Once the product is exported correctly, the unwrapping can be started from within SNAP, by calling the Snaphu-unwrapping operator (under Radar > Interferometric > Unwrapping).
- As an input product, select the subset which was created before the export from the drop-down menu.
- For the output folder in the Processing Parameters tab you can activate Display execution output and select the folder which was created during the export (20251119_20251125_split_Orb_Stack_esd_ifg_deb_flt_subset).
- To start the unwrapping, click Run. SNAP then sends the command stored in snaphu.conf to the snaphu.exe which creates the raster image belonging to the unwrapped phase metadata.

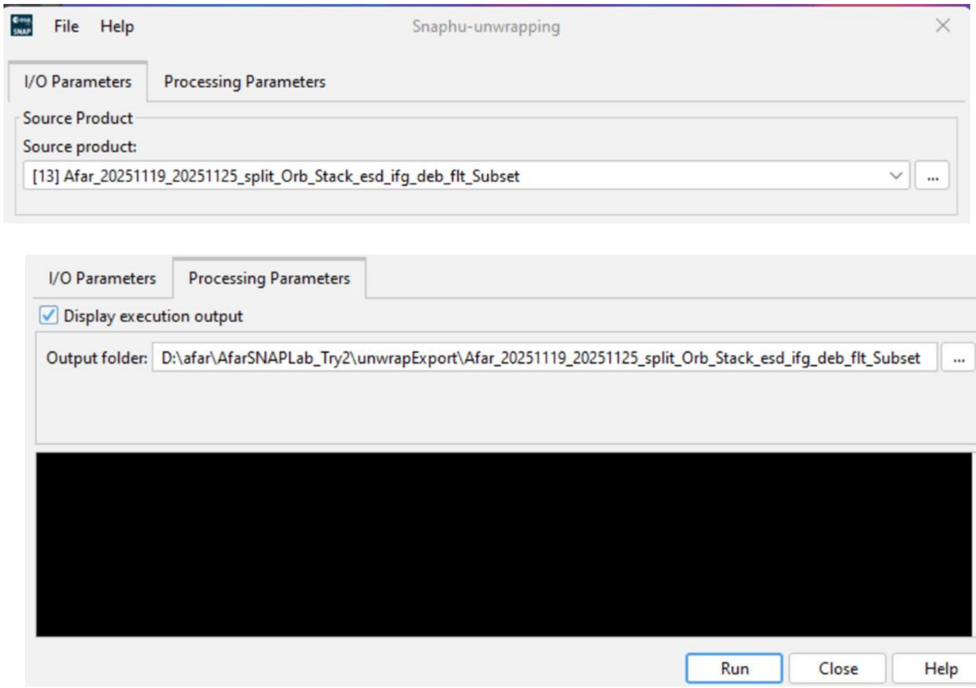
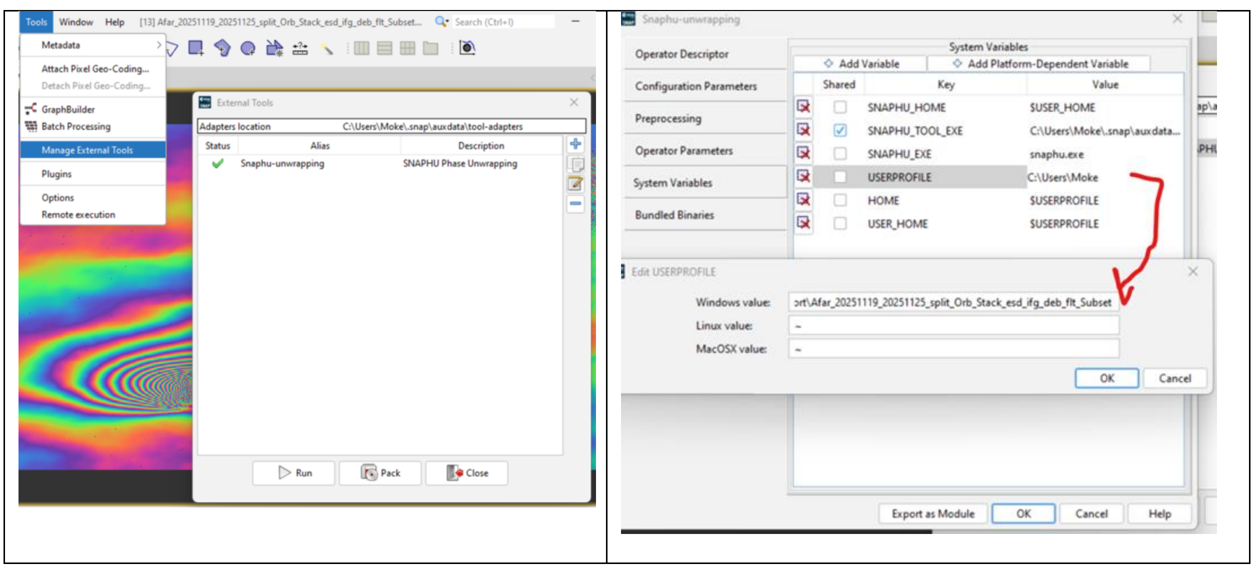


Figure 21: Unwrapping with the snaphu plugin

Note: If the tool is not executed by clicking Run, go to Manage External Tools in the Tools menu. Select “Snaphu-unwrapping” and click Edit the selected operator to open the configuration and enter the export directory in the variable USERPROFILE under the System Variables tab. Confirm with OK and start the Snaphu-unwrapping operator again.



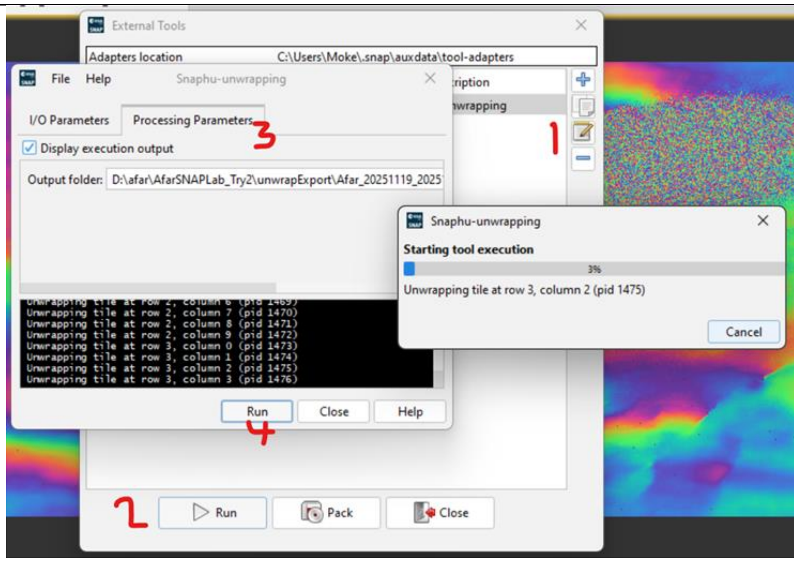


Figure 22: System variables of the snaphu plugin

Note that unwrapping can take a considerable amount of time. To break the raster into smaller chunks, snaphu divides it by standard into 10 rows and 10 columns, leading to a total of 100 tiles to be unwrapped. These tiles are merged afterwards based on the selected pixel overlap (here 200 pixels). A progress bar will inform you about the remaining time (Figure 23). The calculation is completed when the progress reaches 100% and the output states “Finished tool execution in … seconds”. You can then close the operator.

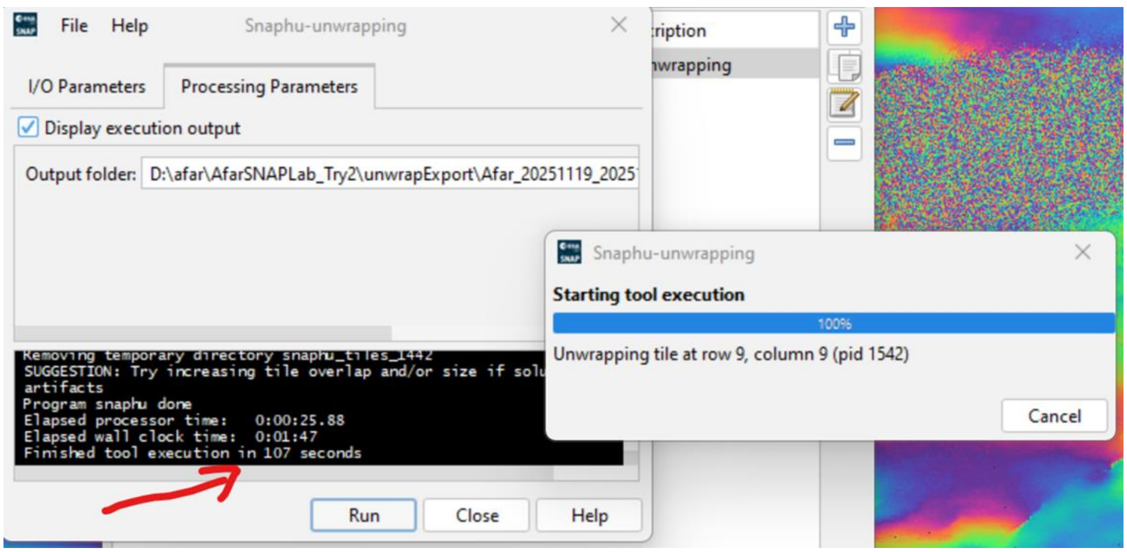


Figure 23: Progress of the unwrapping

Alternative option: If the unwrapping does not work with the plugin from within SNAP, snaphu can also be started from the command line. It can be downloaded from the following page which also contains instruction for its manual execution: https://step.esa.int/main/third-party-plugins-2/snaphu/

3.4.3. Import

The opened unwrapped phase does not have any metadata or geocoding. The Snaphu import (under Radar >Interferometric > Unwrapping) converts the it back into the BEAM DIMAP format and adds the required metadata from on the wrapped phase product as they have the same geometry. The Snaphu import needs the following tabs:

- 1-Read-Phase: Here, you select the subset of the interferogram (before the export) from the drop- down menu.
- 2-Read-Unwrapped-Phase: Select the icon to open a file menu and navigate to the export directory. Select the *.hdr file of the unwrapped phase (here UnwPhase_ifg_[…].hdr). Note: The error message will then vanish if you proceed to the next tab.
- 3-SnaphuImport: Leave the option “Do NOT save Wrapped interferogram in the target product” unchecked, because it is required in the later step.
- 4-Write: To store the imported unwrapped band in a separate product (recommended), add ‘_unw’ to the output name
- 20251119_20251125_split_Orb_Stack_esd_ifg_deb_flt_subset_unw
- Click Run.

A new product is added to the Product Explorer which contains the wrapped interferogram (Phase_ifg_IW…), the coherence (coh_IW_...) and the unwrapped phase generated with snaphu (here: Unw_Phase_ifg_20251119_20251125)

Double click on the unwrapped phase to see if the unwrapping was successful. It should be a smooth raster with little variation except for the areas of expected deformation (Figure 24). All fringe patterns are summarized to absolute changes (Figure 19). Unwrapping errors can occur in areas of low coherence and in urban areas with many vertical objects.

If you notice strange grid-like patterns in your unwrapped results, you can change the amount of tile columns and rows in the export (or directly in the snaphu.conf file) and run the unwrapping again. Just make sure

you delete the unwrapped phase (UnwPhase_ifg_[…].img) and the temporary files in the export directory before you restart the unwrapping. Sometimes it is also favorable to switch the statistical cost- mode from DEFO to SMOOTH or the initial method from MCF to MST and compare the outputs.

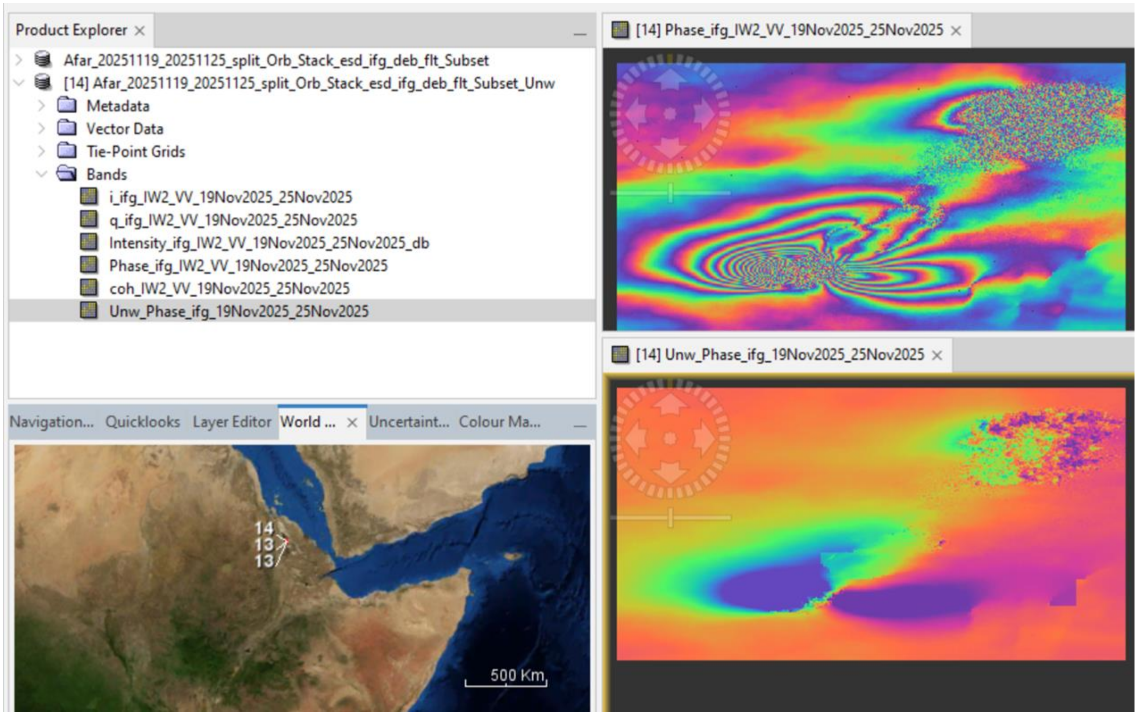
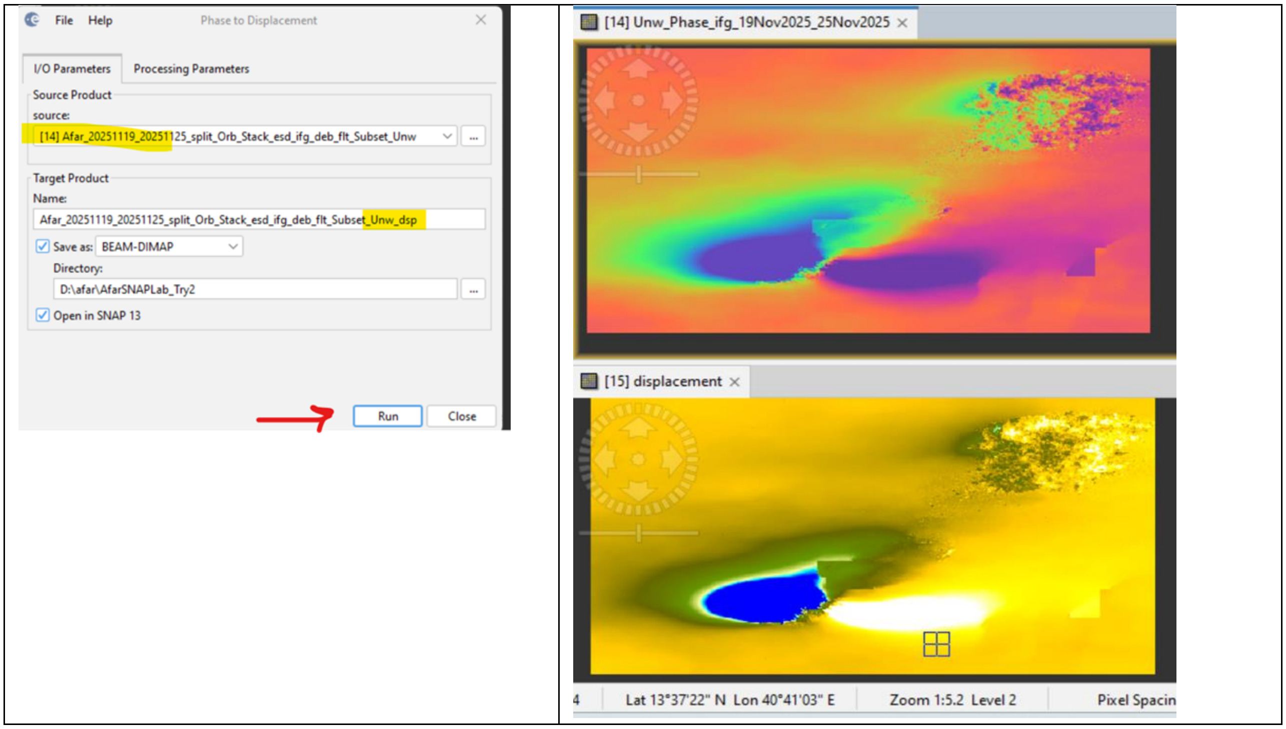


Figure 24: Phase before (top) and after (bottom) unwrapping

3.5.Phase to Displacement

The unwrapped phase is now a continuous raster but not yet a metric measure.

- To convert the radian units into absolute displacements, the Phase to Displacement operator (under Radar > Interferometric > Products) is applied.
- It translates the phase into surface changes along the line-of-sight (LOS) in meters. The LOS is the line between the sensor and a pixel.
- **Accordingly, positive values mean uplift and negative values mean subsidence of the surface.**



The Phase to Displacement operator has no parameters and is applied to the unwrapped phase which was imported in the last step. It produces an output which looks similar to the unwrapped phase (slightly different predefined color ramp), but now each pixel has a metric value indicating its displacement

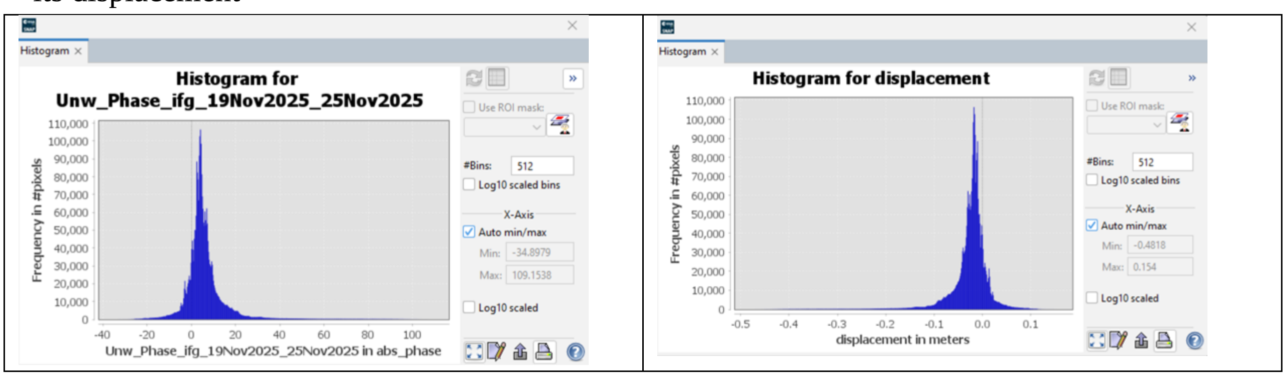


Figure 25: Histogram of unwrapped phase (left) and LOS displacement (right)

3.6.Terrain Correction

Terrain Correction will geocode the image by correcting SAR geometric distortions using a digital elevation model (DEM) and producing a map projected product.
Geocoding converts an image from slant range or ground range geometry into a map coordinate system. Terrain geocoding involves using a Digital Elevation Model (DEM) to correct for inherent geometric distortions, such as foreshortening, layover and shadow. More information on these effects is given in the ESA radar course materials.

- Open the Range Doppler Terrain Correction operator (under Radar > Geometric > Terrain Correction).
- Select the displacement product as an input in the first tab.
- In the Processing Parameters tab, select SRTM 1Sec HGT (AutoDownload) as input DEM.
- If you want to export the data as a KMZ file to view it in Google Earth WGS84 must be selected as Map Projection. It is based on geographic coordinates (latitude and longitude). For the later use in a geographic information system (GIS) a metric map projection, such as UTM (Automatic) could be selected as well.
- Click on Run to start the terrain correction. After completing the terrain correction, SNAP will add the result to product explorer.
- Double click on the output to view the geocoded displacement raster.

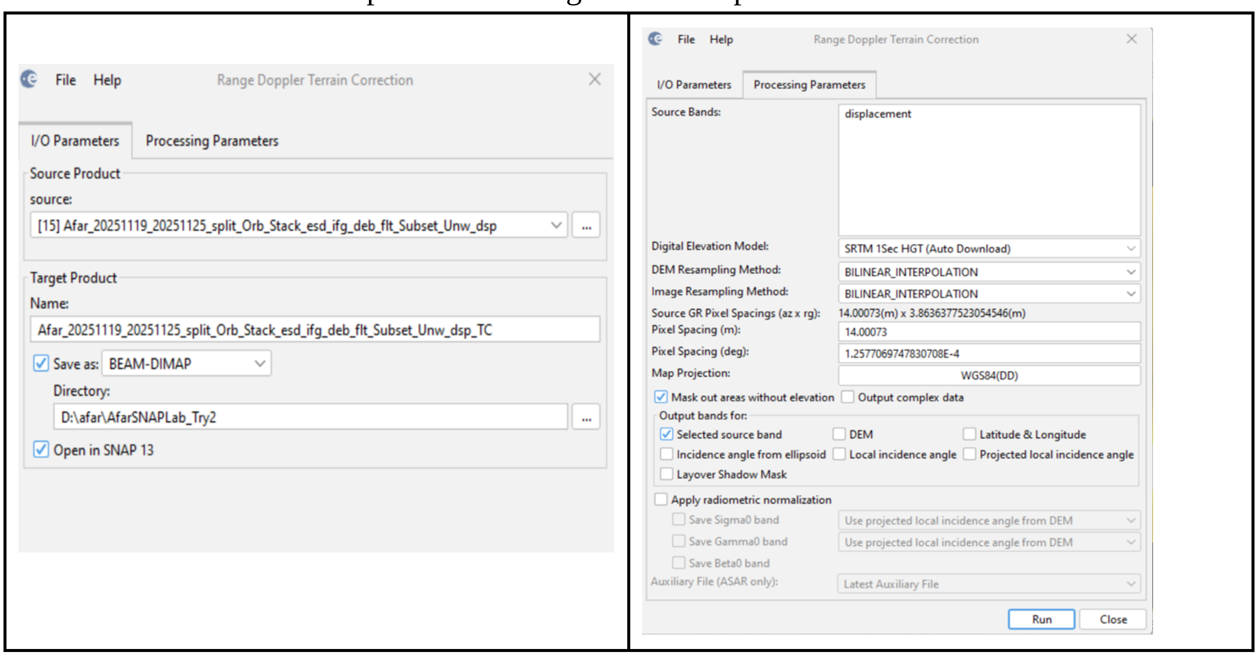


Figure 29: Range Doppler Terrain Correction

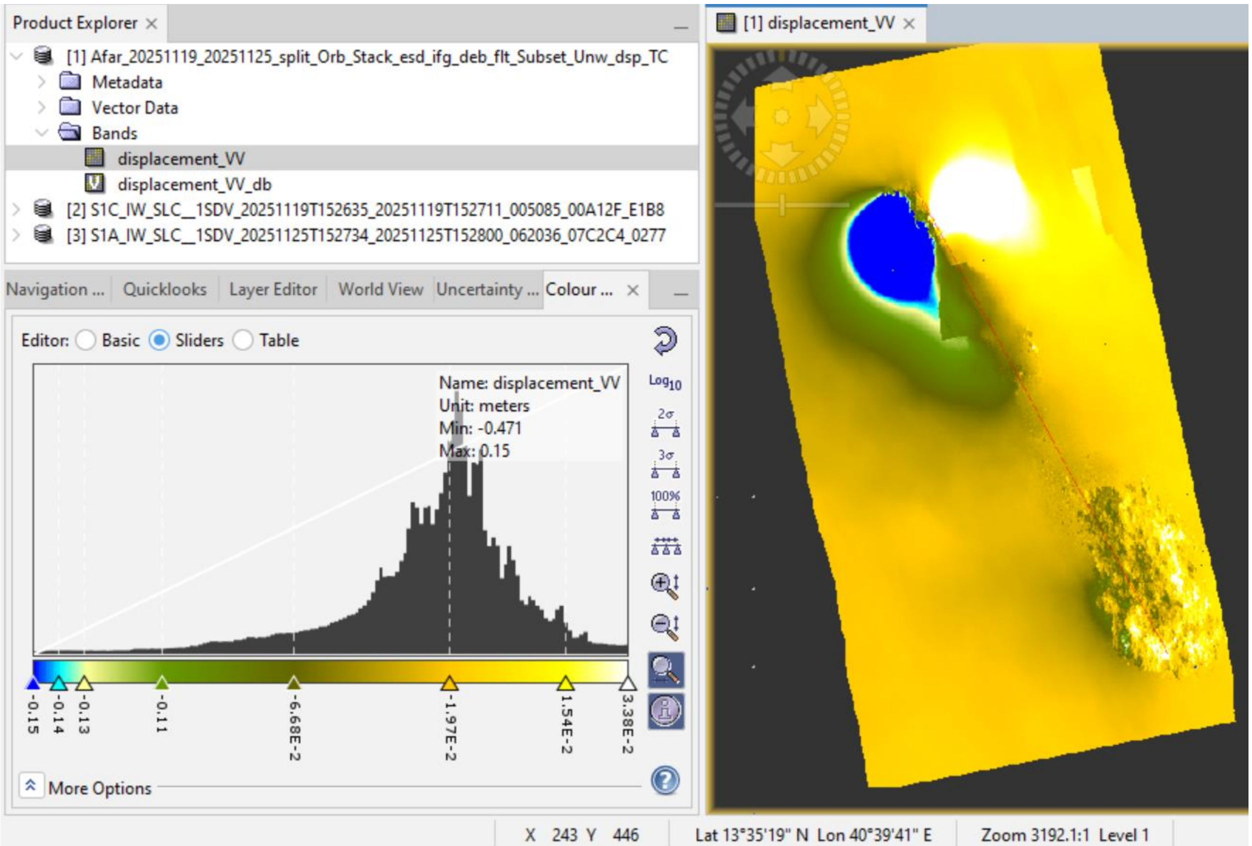


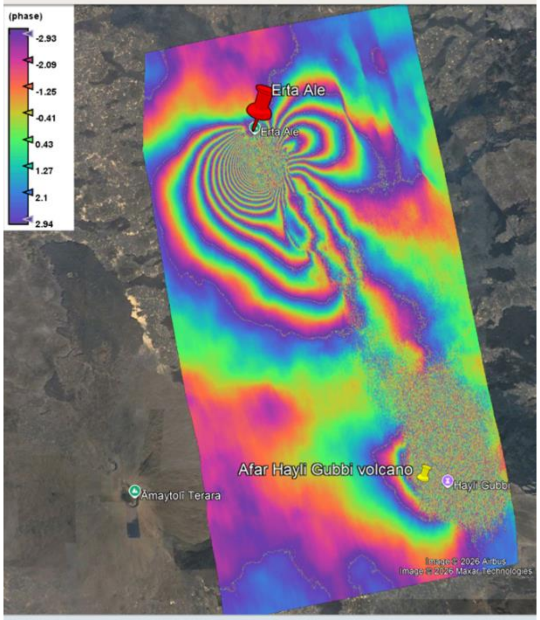
Figure 30: Terrain corrected and geocoded displacement raster

- **Positive values mean uplift and negative values mean subsidence of the surface.**
- **The whiter pixel is, the greater the uplift, The bluer a pixel is, the greater the subsidence.**
- Export to see the movement shown in meters in the legend.

3.7.Export to Google Earth

All geocoded products (projected to WGS84) can be exported as a KMZ file to view them in Google Earth.

- Select a suitable color scale and color ramp in the Color Manipulation tab and then
- Choose File > Export > Other > View As Google Earth KMZ.
- Select an output directory and a suitable filename and confirm with Save.
- Open the kmz file in Google Earth to view the location of the displacement patterns on the satellite image base map (Figure 32).



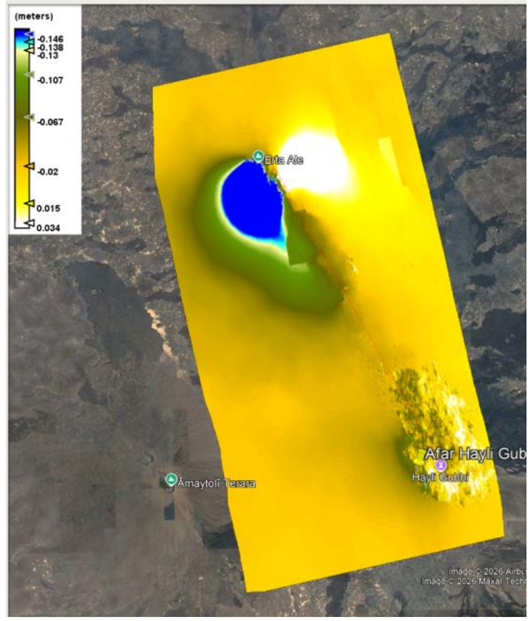


Figure 32: Result exported to Google Earth

3.8.Interpretating Interferogram

This image shows a Differential Interferometric Synthetic Aperture Radar (DInSAR) analysis of ground deformation near the Erta Ale and Hayli Gubbi volcanoes.

Left Panel: Interferometric Phase (Fringes)

- Interferogram shows: The phase difference between two radar images taken at different times.
- Fringe is each full cycle of color (e.g., from purple to purple) represents a specific amount of ground movement along the radar's line of sight (LOS). These patterns(fringes) represent a full 2π cycle(e.g., from purple to purple) and appear in an interferogram as cycles of arbitrary colors, with each cycle representing half the sensor‟s wavelength.
- **Relative ground movement away/towards radar LOS can be derived by counting the fringes and multiplying by half of the wavelength(Sentinel-1 C band is 5.5cm). The closer the fringes are together, the greater the strain on the ground**.
- Dense Fringes: The concentric rings packed closely together near Erta Ale indicate a high gradient of rapid ground displacement.

Right Panel: Displacement Map (Meters)

- Image shows the Unwraped displacement phase: The phase data converted into actual physical measurements of vertical/LOS ground movement.

- **Blue/Green Zone: Represents negative values down to -0.146 meters(-14.6 centimeters), indicating significant ground subsidence or sinking near the Erta Ale crater.**
- **White/Bright Zone: Represents positive values up to positive 0.034 meters(3.4 centimeters), indicating ground uplift or inflation adjacent to the sinking area.**
- **Yellow Zone: Represents stable areas with near-zero displacement.**

The difference between the color bar sliders(figure 32 right) and the actual metadata(figure 30): values comes down to how the software handles data scaling and outliers.

- The Metadata True min max Range(figure 30): The text window shows a true minimum value of -0.471 meters and a maximum of 0.15 meters. This represents the absolute highest and lowest pixel values in the entire unwrapped data layer; **which means some areas show uplift upto 15 Cm and subsidence to 47.1 cm.**
- The Color Bar Scale(figure 32 right): slider scale runs from -0.15 meters to 0.0338 meters (3.38E-2).The software automatically clips or truncates the color stretch to fit the main bulk of the histogram data. This prevents a few extreme, noisy pixels (outliers) from compressing the color variation across the rest of the map. It is recommended to do further reading on interpreting interferogram maps for better analysis and interpretation.

3.9.Suggestions for further analyses

- This tutorial showed how to derive surface displacement from a single image pair of SLC radar data. However, it remains unclear how well the result represents the real conditions. This can only be achieved by external reference data.
- Moreover, to minimize the impact of atmospheric phase disturbances and unwrapping errors, a series of images should be analyzed to get a more reliable result. All steps in this tutorial can be repeated for each consecutive image pair, for instance doing interferometry for images 1 and 2,then 2 and 3, then 3 and 4, and so on.
- By averaging displacements of several dates, the impact of atmospheric disturbances and unwrapping errors is reduced, and actual displacement patterns become more evident.

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